

MOHAMED AHMED ROMY

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PROFESSIONAL SUMMARY

- High-caliber Mechatronics Engineering student with a 3.6/4.0 CGPA, recognized for technical rigor, multi-disciplinary innovation, and hardware-software co-design.
- Dual-certified SOLIDWORKS Professional (CSWP) & Associate (CSWA), alongside 3DEXPERIENCE 3DSwymer credentialing, with advanced proficiency in CAD modeling, DFM/DFA, and FEA analysis.
- Expertise in architecting distributed automation ecosystems and web services utilizing Django 5.2, Asynchronous Task Queues (Celery/Redis), and Real-time WebSockets.
- Proven industrial and automotive competence spanning Siemens S7-1200/1500 PLCs, CAN-bus ECU diagnostics (ManTrac/Caterpillar), classic control systems, and modern vehicle electronic architectures.
- AI Integration Innovator with a demonstrated ability to integrate Generative AI (Groq, Google GenAI) into technical and industrial diagnostics pipelines.

EDUCATION

Bachelor of Mechatronics Engineering

Zagazig National University

Cumulative GPA: 3.6 / 4.0

2023 – Present

Egypt

- Technical Specialization:** Advanced Robotics & Kinematics, Control Systems Engineering (PID/LQR/Infinity), Embedded Systems (AVR/ARM), Machine Design, Mechanical Vibrations, and Finite Element Analysis (FEA).

TECHNICAL PROFICIENCIES

- Mechanical Design & CAD:** SOLIDWORKS (CSWP/CSWA), 3DEXPERIENCE Platform, DFM/DFA, Stress/Thermal Analysis (FEA), CFD, Solar PV Design (PVSyst).
- Software Architecture & Development:** Python (Django 5.2, Django Rest Framework, Channels/WebSockets, Celery, Redis, PostgreSQL), C++, Next.js, TypeScript, Tailwind CSS.
- Industrial Automation & Automotive:** Siemens TIA Portal, S7-1200/1500 PLCs, SCL/STL Programming, HMI/SCADA, Classic Control (Ministry of Industry), CAN-bus Diagnostics, Automotive Electrical Systems.
- AI & Data Science:** LLM Integration (Groq, Google GenAI), Pandas, NumPy, Python Automation (Requests, BeautifulSoup).
- DevOps & Tools:** Linux (WSL/Ubuntu), Bash Scripting, Git/GitHub, Vercel, Antigravity IDE.

PROFESSIONAL EXPERIENCE & TRAINING

C++ Programming Intern

CodeAlpha

March 2025 – April 2025

Remote

- Developed robust C++ software modules, focusing on clean architecture, pointer manipulation, object-oriented principles, and algorithmic problem-solving.

Industrial Maintenance & Diagnostics Intern
ManTrac (Official Caterpillar Dealer)

July 2025
Egypt

- Conducted advanced hardware-level diagnostics on Caterpillar engine control units (ECUs), interpreting CAN-bus data and analyzing heavy machinery electrical fault logs.

Advanced PLC Specialist Training (Siemens S7-1500)
DON-BOSCO Institute

January 2025
Cairo, Egypt

- Engineered high-level automation logic for synchronized multi-axis production lines using SCL and STL.
- Implemented complex PID control loops for temperature and pressure regulation in industrial manufacturing simulations.

Systems Engineering & Control Intern
HA Consulting Group

Summer 2024
Cairo, Egypt

- Developed professional technical schematics for industrial control panels, ensuring strict compliance with IEC/NEC standards.
- Gained intensive hands-on training in Classic Control, Siemens PLC Basic Programming, and AC/DC Motor Drives.

Automotive Mechanical & Electrical Diploma Training
Engovation & InstaFix

2024
Egypt

- Completed comprehensive practical training in modern vehicle electronic control units (ECUs), sensor/actuator diagnostics, and automotive electrical wiring systems.

STRATEGIC TECHNICAL PROJECTS

Unified Mechatronics System Platform (UMSP) | znue-portal.me

2025

- Built a high-availability multidisciplinary portal utilizing Django 5.2 with Redis/Celery background task queues for asynchronous operations.
- Developed an integrated RAG system utilizing Groq and Google GenAI for instant technical data analysis and intelligent system querying.

CERTIFICATIONS & LANGUAGES

- **CSWP & CSWA:** Certified SOLIDWORKS Professional & Associate - Mechanical Design
- **3DSwymer Associate:** Certified 3DEXPERIENCE Platform Credential (Dassault Systèmes)
- **Solar Energy Systems Specialist:** Egyptian Engineering Syndicate & Engovation Academy
- **Classic Control Certificate:** Ministry of Industry, Egypt
- **Cable Manufacturing Technology:** El Sewedy Industrial Training Facility
- **Build with AI (Masr Edition):** Google for Developers & ITI
- **Languages:** Arabic (Native), English (Upper-Intermediate / B2)